

Kanakam SD

(814) 626-9967

ksdimple.967@gmail.com

SUMMARY

- Developed and optimized Java-based enterprise solutions using **Spring Boot**, **Spring Framework**, and **Hibernate**, tackling computational challenges by implementing high-performance algorithms and leveraging object-relational mapping for efficient database operations.
- Architected and deployed scalable **microservices** using **Spring Boot**, **REST APIs**, and **Apache CXF**. This approach increased deployment flexibility, reduced application coupling, and enabled efficient scaling, supporting a 50% increase in traffic with minimal infrastructure changes.
- Designed, implemented, and consumed **RESTful** and **SOAP** web services, using **WSDL**, **UDDI**, and **Apache CXF** for seamless integration across distributed systems. Ensured high availability and reliable data exchange in mission-critical applications.
- Worked extensively with **Spring** modules, including **Spring Security** and **Spring Data**, for end-to-end enterprise application development. Implemented authentication, authorization, and data persistence mechanisms, enabling secure and efficient data access across the application.
- Experience with developing responsive, cross-browser web interfaces using HTML5, CSS3, Bootstrap, SASS, SCSS, etc, incorporating JavaScript, TypeScript, jQuery, AJAX, and JSON to support dynamic, high-performance web applications that enhanced user engagement and reduced page load times.
- Proficiency in enterprise-grade **React** applications with **Redux**, focusing on component-based design and centralized state management to improve application performance and maintainability.
- Hands-on experience with single-page applications using **Angular** and **AngularJS**, employing component-based architectures to enhance application scalability and performance.
- Strong knowledge of **JavaScript**, **TypeScript**, and **ES6+** capabilities for developing type-safe, high-performance web solutions, minimizing runtime errors, and improving code maintainability.
- Led the design and implementation of complex software systems using **OOP** principles, ensuring modular, extensible code architectures. Actively applied **encapsulation**, **inheritance**, and **polymorphism** to build maintainable, scalable applications with clear object hierarchies and reusable components across multiple projects.
- Applied **inheritance** and **polymorphism** strategies in large-scale systems, resulting in significant reductions in code duplication, improved system flexibility, and enhanced maintainability. Managed to extend functionality with minimal changes, ensuring long-term system scalability and reusability.
- Utilized **enterprise design patterns** such as **Singleton**, **Factory**, **Observer**, **MVC**, and **Microservices** to solve complex architectural challenges. Delivered highly flexible, decoupled systems that improved maintainability, allowing for easier updates and enhancements without major system overhauls.
- Led the development and execution of robust test strategies using frameworks like **JUnit**, **Jasmine**, **Karma**, **Jest**, **Mockito**, and **Selenium**, ensuring comprehensive front-end and back-end validation. Reduced post-release bugs through strategic test coverage and continuous integration.
- Leveraged **Java** features like **Streams**, **Lambdas**, and **Functional Programming** to write highly efficient, concise, and readable code, improving both the performance and maintainability of enterprise-level applications. Reduced boilerplate code and enhanced parallel processing capabilities, optimizing key application processes.
- Integrated **dependency injection (DI)** and **inversion of control (IoC)** in complex systems, leading to more modular, loosely coupled applications and enabling easier unit testing and more maintainable code. Successfully reduced system complexity and improved the flexibility of service configurations.
- Optimized database interactions using **Hibernate ORM**, **PL/SQL**, **MySQL**, **MongoDB**, **Redis**, and **Cassandra**, improving performance and ensuring high availability in distributed environments.
- Implemented robust authentication and authorization mechanisms using **JWT** and **OAuth**, enhancing security for both internal services and third-party integrations. Ensured compliance with industry standards and improved user trust and data protection.
- Designed and implemented efficient **distributed messaging** systems using **Kafka**, **RabbitMQ**, **ActiveMQ**, and **JMS**. These systems facilitated decoupled, high-throughput communication between microservices, reducing latency and improving system reliability.

- Developed comprehensive test suites using **JUnit**, **Mockito**, **TestNG**, **Selenium**, **JMeter**, and **Postman**, increasing code quality, automating repetitive tasks, and ensuring stable application performance under load.
- Conducted rigorous **API testing** with tools like **Postman**, **Firebug**, and **SOAP UI**, ensuring reliable and secure integrations between third-party services and internal systems, resulting in fewer system outages and seamless end-user experiences.
- Led build process optimizations with **Maven**, **Gradle**, and **NPM**, streamlining configurations and enhancing development speed while reducing errors.
- Managed **Git workflows** with **Git**, **GitHub**, **Bitbucket**, and **SVN**, implementing branching strategies, pull requests, and code review processes that improved collaboration across distributed teams and ensured consistent code quality throughout the development lifecycle.
- Designed **CI/CD pipelines** with **Jenkins**, **Bamboo**, and **GitHub Actions**, automating deployments and ensuring rapid, consistent delivery with minimal downtime.
- Led the implementation of **Infrastructure-as-Code (IaC)** with **Terraform**, **Ansible**, and **Chef**, enabling repeatable and automated environment provisioning. This approach increased deployment speed and consistency across development, staging, and production environments.
- Architected **microservices** containerization with **Docker** and **Kubernetes**, ensuring reliable deployments and optimizing infrastructure costs.
- Developed multi-region, scalable **cloud infrastructure** on **AWS** and **Azure**, using services such as **EC2**, **S3**, **Lambda**, **VPC**, **IAM**, **EBS**, **ELB**, **Route 53**, **CloudFormation**, and **CloudFront** to ensure high availability and disaster recovery.
- Implemented advanced **cloud security protocols** to secure sensitive enterprise data and ensure compliance with industry regulations. Led efforts to improve overall system resilience against potential security breaches and vulnerabilities.
- Applied **cloud resource optimization strategies**, utilizing **serverless architecture** and dynamic scaling, which reduced operational costs while improving application performance.
- Led **Agile** and **Scrum** initiatives using tools like **JIRA**, **Bugzilla**, and **Mantis Bug Tracker**. This approach significantly improved project tracking, and streamlined team workflows.

TECHNICAL SKILLS

Languages	Java, C, C++, Python, Shell Scripting, Scala, XML, HTML, HTML5
Java Technologies	Spring Boot, Spring MVC, Spring Framework, Hibernate, Servlets, JVM Tuning, JDBC, JSP, JSTL, Multi-threading, Spring Security, Apache CXF, Microservices
Performance Optimization	Kafka, RabbitMQ, Redis, JMeter, JVM Profiling, Garbage Collection, Thread Dumps, Circuit Breakers, Load Balancers
Web Services & APIs	RESTful APIs, SOAP, Apache CXF, Microservices, Event-Driven Architecture, Responsive web Design,
JavaScript Technologies	React, Angular, AngularJS, Node.js, Express.js
Database	Oracle, PL/SQL, MySQL, PostgreSQL, MongoDB, DynamoDB, Query Optimization
Testing frameworks	JUnit, Jasmine, Jest, Karma, TestNG, Mockito, JMeter, Selenium, Protractor
Web/App Server	Tomcat, WebSphere, WebLogic, JBoss, Nginx
OS	Windows, Mac OS, LINUX, UNIX
Devops & CI/CD	Maven, Gradle, NPM, Docker, Kubernetes, Jenkins, Terraform, Ansible, GitHub Actions
Cloud Platforms	AWS, Microsoft Azure, Google Cloud Platform (GCP)

Bug Tracking & Reporting	JIRA, Bugzilla, Mantis Bug Tracker
Continuous Integration Tools	Jenkins, Bamboo, Github Actions
Monitoring & Logging	Nagios, Splunk, CloudWatch, ELK Stack
Version Control	Git, GitHub, Bitbucket, SVN

WORK EXPERIENCE

Wells Fargo, Remote

Jan 2024- Feb 2025

Java Full Stack Developer

Description:

At Wells Fargo, a leader in digital banking and investment solutions, I contributed to the modernization of a large-scale enterprise platform within the digital banking and investment team. As a Senior Java Full Stack Developer, I played a key role in driving platform development, leveraging cutting-edge technologies to enhance performance and deliver innovative solutions for financial institutions.

Responsibilities:

- Led redesign of a legacy monolithic application into microservices by implementing **CQRS** with event sourcing, improving response times while maintaining data consistency across bounded contexts.
- Built an event-driven notification system using **Spring Boot** and **Apache Kafka**, implementing exactly-once delivery guarantees and sophisticated error handling through dead-letter queues.
- Architected a versioned API gateway using **Spring Cloud Gateway**, implementing adaptive rate limiting and circuit breakers to enhance system resilience and maintain high availability.
- Implemented resiliency patterns such as Circuit Breakers, **Load Balancers**, and Failover mechanisms using **Spring Cloud**, ensuring 99.9% system uptime during peak traffic.
- Identified and resolved performance bottlenecks in a high-volume digital banking platform, reducing latency through **JVM tuning**, garbage collection optimization, and **multi-threading** strategies.
- Designed and implemented event-driven microservices using **Apache Kafka** and **RabbitMQ**, enabling high-throughput, low-latency communication between services in a **distributed** banking platform.
- Optimized **PostgreSQL** performance by implementing indexing strategies, query optimization, and partitioning techniques, reducing query execution times by and improving overall system performance.
- Developed advanced **SQL** procedures to automate the transformation and validation of JSON data ingested from multiple banking systems, enabling seamless integration with Oracle tables and improving data processing efficiency.
- Developed **shell scripts** to automate deployment, monitoring, tasks on **Linux** servers, reducing manual effort and improving reliability through log rotation, and performance monitoring.
- Built a comprehensive design system library using **CSS Grid/Flexbox** and SCSS modules, creating reusable components that standardized the UI/UX across mobile and desktop platforms while dramatically reducing CSS complexity.
- Designed a real-time collaborative dashboard using **React** and **Redux Toolkit**, implementing custom middleware for concurrent edit resolution and optimistic updates with WebSocket communication.
- Engineered an advanced code-splitting strategy with **TypeScript** and **Webpack**, implementing granular chunking and tree-shaking to optimize bundle sizes for improved initial page loads and caching.
- Designed a flexible plugin architecture using **TypeScript decorators** and dependency injection, creating an extensible platform for third-party developers to seamlessly integrate custom functionality.
- Implemented domain-driven design patterns across microservices, using **event sourcing** and CQRS to handle complex business workflows while maintaining service autonomy and data consistency.
- Created a comprehensive **E2E testing** framework combining **Cypress** with custom test generators, focusing on critical user paths while minimizing test maintenance through smart selectors.
- Developed a high-throughput payment processing service utilizing **Java virtual threads** and reactive programming, ensuring consistent low latency even during peak load periods.

- Modernized legacy SOAP services through a **facade pattern API adapter**, enabling gradual migration while preserving existing client integrations through backward compatibility.
- Created a multi-tenant authentication system using **JWT** with dynamic key rotation and **OAuth2 resource server**, supporting organizational isolation while maintaining a seamless user experience.
- Optimized database access patterns using **Hibernate** with custom batch processing and second-level cache strategies, significantly improving query performance for high-traffic endpoints.
- Designed a real-time analytics pipeline using **Kafka Streams** and **ksqlDB**, implementing window aggregations and anomaly detection for large-scale data processing with minimal latency.
- Built a **contract-testing** framework using **Pact** with custom validators, automating API test generation from OpenAPI specifications to ensure comprehensive coverage of edge cases.
- Developed advanced test suites using **JUnit** and **TestContainers**, focusing on mutation testing to ensure robust coverage of core business logic.
- Troubleshooted and resolved **Linux** server issues, including performance bottlenecks, memory leaks, and network connectivity problems, ensuring uptime for critical systems.
- Monitored Linux server performance using tools like Nagios and **CloudWatch**, identifying and resolving issues before they impacted end-users.
- Implemented blue-green deployment automation using **Jenkins pipelines** and **Spinnaker**, enabling frequent deployments with automated canary analysis and rollback capabilities.
- Established trunk-based development practices with feature flags using **LaunchDarkly**, streamlining the development workflow and enabling controlled feature releases.
- Led containerization initiative using **Docker**, creating optimized images and deployment strategies that significantly improved application portability and startup times.
- Designed auto-scaling **Kubernetes** infrastructure with custom metric-based horizontal pod autoscaling, ensuring optimal resource utilization during varying load conditions.
- Created multi-region **AWS** infrastructure using Transit Gateway and **Route53 failover**, implementing comprehensive disaster recovery capabilities for business continuity.
- Developed a serverless image processing pipeline using **AWS Lambda** and S3, implementing automatic format optimization and responsive image generation.
- Established **GitOps** workflows using **ArgoCD** and Helm, enabling declarative infrastructure management and self-service capabilities for development teams.
- Built a centralized observability platform using **ELK stack** with custom Beats agents, enabling rapid incident response through correlated monitoring of logs, metrics, and traces.
- Led an agile development team while implementing automated acceptance criteria and WIP limits, focusing on sustainable delivery and continuous improvement.
- Established **architecture decision records (ADRs)** process, documenting key technical decisions and their context to facilitate future system evolution and onboarding.
- Created an internal developer platform with custom CLI tools and self-service portals, streamlining environment setup and promoting development best practices.
- Implemented comprehensive security measures leading to **SOC2 certification**, including automated security scanning and continuous compliance monitoring.
- Developed custom static analysis rules using **SonarQube**, enforcing code quality standards and preventing technical debt accumulation through automated checks.

Environment: Java, Spring Boot, RESTful APIs, SOAP, JWT, OAuth2, Hibernate, Kafka, RabbitMQ, HTML5, CSS3, Bootstrap, React, Redux, JavaScript, TypeScript, Factory, Strategy, Observer, Enzyme, Karma, JUnit, Jenkins, Maven, Git, Docker, Kubernetes, AWS, S3, Lambda, EC2, ELK stack, Jira, SonarQube

Smarsh, Bengaluru

Feb 2020 - Dec 2022

Full Stack Developer

Description: Smarsh/DigitalSafe provides compliance and archiving solutions for regulated industries. I worked with the Reporting Team, building scalable backend systems for transaction processing and real-time

data handling. My role involved developing secure APIs, optimizing database performance, automating deployments, and improving system reliability.

Responsibilities:

- Developed a distributed transaction processing system using **Spring Boot** microservices with **RabbitMQ** event queues, implementing Dead Letter Exchanges (DLQ) and message acknowledgment patterns for handling failed transactions and retries.
- Implemented custom **OAuth2** resource server with **JWT** tokens, building role-based security using **@PreAuthorize** annotations and custom SpEL expressions for complex authorization rules.
- Built dynamic form validation system using **Angular** reactive forms with custom validators, async validators, and error state matches, integrated with backend validation using **Spring's @Valid**.
- Created a caching abstraction layer using **Spring Cache** with **Redis**, implementing cache-aside pattern and custom key generators for handling complex object hierarchies, reducing redundant database calls for frequently accessed data.
- Designed **Kubernetes** deployment configurations with horizontal pod autoscaling based on custom metrics, implementing readiness/liveness probes and configuring ingress controllers with SSL termination.
- Built custom **Spring Data JPA** specifications for dynamic query building, implementing composite specifications with AND/OR conditions and handling nested entity relationships with fetch joins for optimized queries.
- Implemented WebSocket communication using **Spring's STOMP** protocol support, handling real-time bidirectional data flow with message brokers and implementing custom channel interceptors for session management.
- Created **Azure DevOps** pipelines with multi-stage deployments, implementing environment-specific variable groups and secure secret management using **Azure Key Vault** integration.
- Built RESTful APIs using **Oracle** REST Data Services (ORDS) to expose JSON data stored in Oracle tables, enabling seamless integration with frontend applications and third-party systems.
- Designed and implemented a dynamic reporting system using **JSON data** stored in Oracle tables, leveraging PL/SQL procedures to transform JSON into relational format and integrating with JasperSoft for real-time data visualization and customizable business reports.
- Developed custom **Angular** structural directives for handling complex DOM manipulations and conditional rendering, implementing micro syntax parsing and context injection.
- Built **Terraform** modules for Azure infrastructure provisioning, implementing remote state management with Azure Storage and state locking mechanisms.
- Created custom annotation processors in **Spring Boot** for generating boilerplate code during compile-time, reducing repetitive code patterns in DTO mappings and validation.
- Implemented event sourcing pattern using **Spring Cloud Stream** and **RabbitMQ**, maintaining audit trails and enabling event replay capabilities for data recovery.
- Built custom **Gradle plugins** for automating deployment tasks, implementing incremental build support and custom task dependencies for optimized build processes.
- Developed **Selenium** test framework with Page Object Model pattern, implementing custom wait conditions and handling dynamic elements with JavaScript execution.
- Created reusable **TypeScript decorators** for method timing, logging, and error handling, implementing metadata reflection for runtime type information.
- Built custom **Spring Boot actuator** endpoints for application monitoring, implementing health indicators and custom metrics collection for **Azure Application Insights**.
- Implemented database sharding strategy using **Spring's AbstractRoutingDataSource**, creating custom routing keys based on tenant information for multi-tenant applications.
- Developed **Jenkins-shared** libraries for common pipeline tasks, implementing custom steps for automated testing and deployment processes.
- Created custom **Angular HTTP interceptors** for handling authentication, request/response transformation, and global error handling with retry mechanisms.
- Built **SonarQube** quality gates with custom rules and metrics, implementing automated code review processes with pull request decoration.
- Implemented custom **JUnit extensions** for testing **Spring Boot** applications, creating test fixtures and custom annotations for test configuration.

- Created **Docker** multi-stage builds for optimizing image sizes, implementing layer caching strategies, and security scanning with vulnerability assessment.
- Developed custom **Thymeleaf dialect** for complex server-side rendering requirements, implementing custom attribute processors and expression objects.
- Built API versioning strategy using custom request mappings and content negotiation in **Spring MVC**, handling backward compatibility for legacy clients.
- Implemented custom **Hibernate** user types for handling complex domain objects, creating custom type descriptors and SQL type mappings.

Environment: HTML5, CSS3, Ajax, Bootstrap, Angular, JavaScript, TypeScript, Selenium, Java, RESTful APIs, SOAP, Spring Boot, Hibernate, JWT, OAuth2, RabbitMQ, JUnit, Maven, Git, Docker, Kubernetes, AWS, Jenkins, Jira, Terraform

MicroFocus, Remote

Feb 2017 - Jan 2020

Jr. Software Engineer

Description:

At MicroFocus, a leading enterprise software provider, worked on a project to develop a single-page application with advanced filtering and payment gateway integration. As a Junior. Software Engineer focused on building frontend components, implementing security features, and assisting with backend API development. Contributed to enhancing system reliability and supporting deployment automation processes.

Responsibilities:

- **Developed** and maintained a single-page application using **React**, ensuring seamless user experience across all devices with optimized rendering and responsiveness.
- **Implemented** an **Advanced Filtering System** that enables users to create nested, complex queries from the frontend, dynamically generating corresponding **RESTful APIs** and modifying the **backend** for enhanced query execution.
- **Integrated TypeScript** into the project for static type checking, significantly improving code maintainability and reducing runtime errors.
- **Designed and developed** reusable, modular **UI components** using **React, Redux, and SCSS**, establishing a consistent and scalable design system across the product.
- **Engineered** and **optimized** robust **RESTful APIs** using **Java, Spring Boot, and Hibernate**, ensuring seamless data exchange and backend integration.
- **Integrated** multiple **payment gateway APIs** using secure protocols, ensuring encrypted and efficient transaction processing.
- **Enhanced security** by implementing **JWT token authentication and OAuth**, securing API endpoints and ensuring robust authorization mechanisms.
- **Utilized MySQL and PostgreSQL** for structured data storage and optimized database queries, ensuring efficient data retrieval and manipulation.
- **Configured and maintained CI/CD pipelines** using **Jenkins, Docker, and Kubernetes**, automating the build, test, and deployment processes, significantly improving development workflow efficiency.
- **Leveraged AWS services** such as **S3, EC2, and Document DB** for scalable infrastructure, high availability, and efficient data management, contributing to improved system reliability and performance.

Environment : React, TypeScript, Redux, SCSS, Java, Spring Boot, Hibernate, JWT, OAuth, MySQL, PostgreSQL, AWS (S3, EC2, Document DB), Jenkins, Docker, Kubernetes.